

Topic C4: Predicting and identifying reactions and products

C4.1 Predicting chemical reactions

Summary

Models of how substances react and the different types of chemical reactions that can occur enable us to predict the likelihood and outcome of a chemical reaction. The current Periodic Table was developed based on observations of the similarities and differences in the properties of elements. The way that the Periodic Table is arranged into groups and periods reveals the trends and patterns in the behaviour of the elements. The model of atomic structure provides an explanation for trends and patterns in the properties of elements. The arrangement of elements in groups and periods reveals the relationship between observable properties and how electrons are arranged in the atoms of each element.

Underlying knowledge and understanding

Learners should be familiar with the principles underpinning the Mendeleev Periodic Table; the Periodic Table: periods and groups; metals and non-metals; the varying physical and chemical properties of different elements; the chemical properties of metals and non-metals; the chemical properties of metal and

non-metal oxides with respect to acidity and how patterns in reactions can be predicted with reference to the Periodic Table.

Common misconceptions

Learners consider the properties of particles of elements to be the same as the bulk properties of that element. They tend to rely on the continuous matter model rather than the particle model. Learners confuse state changes and dissolving with chemical changes. Also, since the atmosphere is invisible to the eye and learners rely on concrete, visible information, this means they therefore often avoid the role of oxygen in their explanations for open system reactions. Even if the role of oxygen is appreciated, learners do not realise that solid products of an oxidation reaction have more mass than the starting solid.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers. All other statements will be assessed in both Foundation and Higher Tier papers.

Reference	Mathematical learning outcomes	Mathematical skills
CM4.1i	arithmetic computation and ratio when determining empirical formulae, balancing equations	M1a, M1c

Topic content			Opportunities to cover:		Practical suggestions
Learning outcomes		To include	Maths	Working scientifically	
C4.1a	recall the simple properties of Groups 1, 7 and 0	physical and chemical properties		WS1.2a, WS1.4a WS1.4c	Displacement reactions of halogens with halides. (PAG C1)
C4.1b	explain how observed simple properties of Groups 1, 7 and 0 depend on the outer shell of electrons of the atoms and predict properties from given trends down the groups	ease of electron gain or loss; physical and chemical properties			
C4.1c ☒	recall the general properties of transition metals and their compounds and exemplify these by reference to a small number of transition metals	melting point, density, reactivity, formation of coloured ions with different charges and uses as catalysts		WS1.4a	Investigation of transition metals. (PAG C1, PAG C5, PAG C8)
C4.1d	predict possible reactions and probable reactivity of elements from their positions in the Periodic Table			WS1.1b, WS1.2a, WS1.4a	
C4.1e	explain how the reactivity of metals with water or dilute acids is related to the tendency of the metal to form its positive ion		M1a, M1c	WS1.4a	Reaction of metals with water, dilute hydrochloric acid. (PAG C1, PAG C7, PAG C8)
C4.1f	deduce an order of reactivity of metals based on experimental results			WS1.3e, WS2a	Displacement reactions involving metals and metal salts. (PAG C1, PAG C7, PAG C8)